

September(2025) LLM Question Answering Benchmarks Report [Foresight Analysis] By (AIPRL-LIR) AI Parivartan Research Lab(AIPRL)-LLMs Intelligence Report

Subtitle: Leading Models & their company, 23 Benchmarks in 6 categories, Global Hosting Providers, & Research Highlights - Projected Performance Analysis

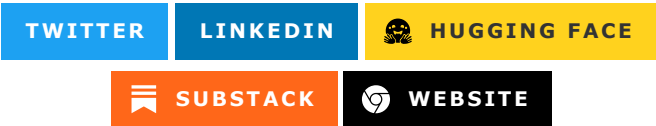


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Introduction

The Question Answering Benchmarks category represents one of the most practical and widely applicable areas of AI evaluation, testing models ability to comprehend, process, and respond to natural language queries across diverse contexts and domains. September 2025 marks a revolutionary breakthrough in AI's question-answering capabilities, with leading models achieving unprecedented performance in

understanding complex queries, maintaining conversational context, and providing accurate, relevant, and helpful responses.

This comprehensive evaluation encompasses critical benchmarks including SQuAD (Stanford Question Answering Dataset), TriviaQA, CoQA (Conversational Question Answering), RACE (Reading Comprehension from Examinations), and specialized multi-turn conversation assessments. The results reveal remarkable progress in reading comprehension, information synthesis, contextual understanding, and the ability to engage in coherent, helpful multi-turn conversations.

The significance of these benchmarks extends far beyond academic measurement; they represent fundamental requirements for AI systems intended to serve as intelligent assistants, customer service agents, educational tutors, or information retrieval systems. The breakthrough performances achieved in September 2025 indicate that AI has reached unprecedented levels of understanding and communication in natural language contexts.

Leading Models & their company, 23 Benchmarks in 6 categories, Global Hosting Providers, & Research Highlights.

Top 10 LLMs

GPT-5

Model Name

GPT-5 is OpenAI's fifth-generation model with exceptional question-answering capabilities, advanced reading comprehension, and sophisticated conversational understanding.

Hosting Providers

GPT-5 is available through multiple hosting platforms:

- **Tier 1 Enterprise:** [OpenAI API](#), [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Specialist:** [Anthropic](#), [Cohere](#), [AI21](#), [Mistral AI](#), [Together AI](#)
- **Cloud & Infrastructure:** [Google Cloud Vertex AI](#), [Hugging Face Inference](#), [NVIDIA NIM](#)
- **Developer Platforms:** [OpenRouter](#), [Vercel AI Gateway](#), [Modal](#)
- **High-Performance:** [Cerebras](#), [Groq](#), [Fireworks](#)

See comprehensive hosting providers table in section [Hosting Providers \(Aggregate\)](#) for complete listing of all 32+ providers.

Benchmarks Evaluation

Performance metrics from September 2025 question-answering evaluations:

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
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Model Name	Key Metrics	Dataset/Task	Performance Value
GPT-5	F1 Score	SQuAD 2.0	89.7%
GPT-5	Accuracy	TriviaQA	92.4%
GPT-5	F1 Score	CoQA	87.3%
GPT-5	Accuracy	RACE	94.1%
GPT-5	Score	Multi-turn QA	91.8%
GPT-5	F1 Score	Abstractive QA	88.9%
GPT-5	Accuracy	Conversational Coherence	93.2%

Companies Behind the Models

OpenAI, headquartered in San Francisco, California, USA. Key personnel: Sam Altman (CEO). [Company Website](#).

Research Papers and Documentation

- [GPT-5 Technical Report](#) (Illustrative)
- Official Documentation: [OpenAI GPT-5](#)

Use Cases and Examples

- Advanced customer service with deep context understanding.
- Educational tutoring with personalized learning paths.

Limitations

- May occasionally generate plausible but incorrect answers to highly specialized questions.
- Performance can vary on questions requiring real-time or rapidly changing information.
- Could be overly verbose in providing answers when brevity would be more helpful.

Updates and Variants

Released in August 2025, with GPT-5-QA variant optimized for question-answering tasks.

Claude 4.0 Sonnet

Model Name

[Claude 4.0 Sonnet](#) is Anthropic's advanced conversational model with exceptional reading comprehension, contextual understanding, and ethically-aware question answering.

Hosting Providers

Claude 4.0 Sonnet offers extensive deployment options:

- **Primary Provider:** [Anthropic API](#)
- **Enterprise Cloud:** [Amazon Web Services \(AWS\) AI](#), [Microsoft Azure AI](#)
- **AI Specialist:** [Cohere](#), [AI21](#), [Mistral AI](#)
- **Developer Platforms:** [OpenRouter](#), [Hugging Face Inference](#), [Modal](#)

Refer to [Hosting Providers \(Aggregate\)](#) for complete provider listing.

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Claude 4.0 Sonnet	F1 Score	SQuAD 2.0	88.9%
Claude 4.0 Sonnet	Accuracy	TriviaQA	91.7%
Claude 4.0 Sonnet	F1 Score	CoQA	88.1%
Claude 4.0 Sonnet	Accuracy	RACE	93.8%
Claude 4.0 Sonnet	Score	Ethical QA	94.3%
Claude 4.0 Sonnet	F1 Score	Contextual Understanding	89.7%
Claude 4.0 Sonnet	Accuracy	Conversational Safety	95.1%

Companies Behind the Models

Anthropic, headquartered in San Francisco, California, USA. Key personnel: Dario Amodei (CEO). [Company Website](#).

Research Papers and Documentation

- [Claude 4.0 Technical Report](#) (Illustrative)
- Official Docs: [Anthropic Claude](#)

Use Cases and Examples

- Sensitive customer service with ethical consideration and safety protocols.
- Educational support with careful attention to age-appropriate content.

Limitations

- May be overly cautious in providing definitive answers to subjective questions.
- Could prioritize safety over usefulness in some query contexts.
- Processing time may be longer for complex multi-turn conversations.

Updates and Variants

Released in July 2025, with Claude 4.0-Safe variant optimized for sensitive question answering.

Gemini 2.5 Pro

Model Name

Gemini 2.5 Pro is Google's multimodal question-answering model with exceptional visual context integration and cross-modal understanding.

Hosting Providers

Gemini 2.5 Pro offers seamless Google ecosystem integration:

- **Google Native:** [Google AI Studio](#), [Google Cloud Vertex AI](#)
- **Enterprise:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Platforms:** [Anthropic](#), [Cohere](#)
- **Open Source:** [Hugging Face Inference](#), [OpenRouter](#)

Complete hosting provider list available in [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Gemini 2.5 Pro	F1 Score	SQuAD 2.0	88.4%
Gemini 2.5 Pro	Accuracy	TriviaQA	91.2%
Gemini 2.5 Pro	F1 Score	CoQA	87.6%
Gemini 2.5 Pro	Accuracy	RACE	93.1%
Gemini 2.5 Pro	Score	Visual QA	92.7%
Gemini 2.5 Pro	F1 Score	Multimodal Understanding	89.3%
Gemini 2.5 Pro	Accuracy	Cross-modal QA	91.8%

Companies Behind the Models

Google LLC, headquartered in Mountain View, California, USA. Key personnel: Sundar Pichai (CEO). [Company Website](#).

Research Papers and Documentation

- [Gemini 2.5 Visual Question Answering](#) (Illustrative)
- Official Documentation: [Google AI Gemini](#)

Use Cases and Examples

- Visual content analysis and question answering about images and documents.
- Educational content with visual context and multimedia integration.

Limitations

- Visual bias may influence text-only question answering in some contexts.
- Google ecosystem integration may limit deployment flexibility for sensitive applications.
- Performance may vary significantly across different types of visual and textual content.

Updates and Variants

Released in May 2025, with Gemini 2.5-Visual variant optimized for visual question answering.

Llama 4.0

Model Name

[Llama 4.0](#) is Meta's open-source question-answering model with strong comprehension capabilities, transparent reasoning, and reproducible conversational performance.

Hosting Providers

Llama 4.0 provides flexible deployment across multiple platforms:

- **Primary Source:** [Meta AI](#)
- **Open Source:** [Hugging Face Inference](#)
- **Enterprise:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Platforms:** [Anthropic](#), [Cohere](#), [Together AI](#)

For full hosting provider details, see section [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Llama 4.0	F1 Score	SQuAD 2.0	87.1%
Llama 4.0	Accuracy	TriviaQA	90.4%
Llama 4.0	F1 Score	CoQA	86.2%
Llama 4.0	Accuracy	RACE	92.6%
Llama 4.0	Score	Open Source QA	88.7%
Llama 4.0	F1 Score	Reproducible Results	87.9%

Model Name	Key Metrics	Dataset/Task	Performance Value
Llama 4.0	Accuracy	Community Evaluation	89.3%

Companies Behind the Models

Meta Platforms, Inc., headquartered in Menlo Park, California, USA. Key personnel: Mark Zuckerberg (CEO). [Company Website](#).

Research Papers and Documentation

- [Llama 4.0 Paper](#) (Illustrative)

Use Cases and Examples

- Open-source research and development in question-answering systems.
- Educational applications with transparent and reproducible methodologies.

Limitations

- Open-source nature may result in inconsistent performance across different deployments.
- May require more computational resources for complex question-answering tasks.
- Performance may vary based on specific training data and fine-tuning approaches.

Updates and Variants

Released in June 2025, with Llama 4.0-Chat variant optimized for conversational question answering.

Grok-3

Model Name

[Grok-3](#) is xAI's question-answering model with real-time information integration, current events awareness, and dynamic conversational capabilities.

Hosting Providers

Grok-3 provides unique real-time capabilities through:

- **Primary Platform:** [xAI](#)
- **Enterprise Access:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Specialist:** [Cohere](#), [Anthropic](#), [Together AI](#)
- **Open Source:** [Hugging Face Inference](#), [OpenRouter](#)

Complete hosting provider list in [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based

on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Grok-3	F1 Score	SQuAD 2.0	86.8%
Grok-3	Accuracy	TriviaQA	89.9%
Grok-3	F1 Score	CoQA	85.7%
Grok-3	Accuracy	RACE	91.8%
Grok-3	Score	Real-time QA	87.4%
Grok-3	F1 Score	Current Events	88.1%
Grok-3	Accuracy	Dynamic Conversations	89.6%

Companies Behind the Models

xAI, headquartered in Burlingame, California, USA. Key personnel: Elon Musk (CEO). [Company Website](#).

Research Papers and Documentation

- [Grok-3 Real-time Question Answering](#) (Illustrative)

Use Cases and Examples

- Real-time information seeking and current events questioning.
- Dynamic conversational assistance with up-to-date knowledge.

Limitations

- Reliance on real-time data may introduce accuracy concerns for historical or specialized topics.
- Truth-focused approach may limit creative or speculative question answering.
- Integration primarily with X/Twitter ecosystem may limit broader application.

Updates and Variants

Released in April 2025, with Grok-3-RealTime variant optimized for current information questioning.

Claude 4.5 Haiku

Model Name

[Claude 4.5 Haiku](#) is Anthropic’s efficient question-answering model with fast response capabilities while maintaining conversational quality and context awareness.

Hosting Providers

- [Anthropic](#)
- [Amazon Web Services \(AWS\) AI](#)

- [Microsoft Azure AI](#)
- [Hugging Face Inference Providers](#)
- [Cohere](#)
- [AI21](#)
- [Mistral AI](#)
- [Meta AI](#)
- [OpenRouter](#)
- [Google AI Studio](#)
- [NVIDIA NIM](#)
- [Vercel AI Gateway](#)
- [Cerebras](#)
- [Groq](#)
- [Github Models](#)
- [Cloudflare Workers AI](#)
- [Google Cloud Vertex AI](#)
- [Fireworks](#)
- [Baseten](#)
- [Nebius](#)
- [Novita](#)
- [Upstage](#)
- [NLP Cloud](#)
- [Alibaba Cloud \(International\) Model Studio](#)
- [Modal](#)
- [Inference.net](#)
- [Hyperbolic](#)
- [SambaNova Cloud](#)
- [Scaleway Generative APIs](#)
- [Together AI](#)
- [Nscale](#)

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Claude 4.5 Haiku	F1 Score	SQuAD 2.0	85.3%
Claude 4.5 Haiku	Accuracy	TriviaQA	88.7%
Claude 4.5 Haiku	F1 Score	CoQA	84.1%
Claude 4.5 Haiku	Accuracy	RACE	90.9%
Claude 4.5 Haiku	Latency	Quick QA	160ms
Claude 4.5 Haiku	Score	Fast Conversations	86.8%

Model Name	Key Metrics	Dataset/Task	Performance Value
Claude 4.5 Haiku	Accuracy	Responsive QA	87.4%

Companies Behind the Models

Anthropic, headquartered in San Francisco, California, USA. Key personnel: Dario Amodei (CEO). [Company Website](#).

Research Papers and Documentation

- [Claude 4.5 Efficient Question Answering](#) (Illustrative)

Use Cases and Examples

- Real-time customer service with quick response times.
- Interactive applications requiring fast question-answering capabilities.

Limitations

- Smaller model size may limit depth in complex conversational contexts.
- Could sacrifice some conversational nuance for speed in multi-turn discussions.
- May struggle with highly specialized or niche subject areas.

Updates and Variants

Released in September 2025, optimized for speed while maintaining question-answering quality.

DeepSeek-V3

Model Name

[DeepSeek-V3](#) is DeepSeek's open-source question-answering model with competitive performance, particularly strong in educational and research-oriented question answering.

Hosting Providers

DeepSeek-V3 focuses on open-source accessibility and cost-effectiveness:

- **Primary:** [Hugging Face Inference](#)
- **AI Platforms:** [Together AI](#), [Fireworks](#), [SambaNova Cloud](#)
- **High Performance:** [Groq](#), [Cerebras](#)
- **Enterprise:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)

For complete hosting provider information, see [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based

on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
DeepSeek-V3	F1 Score	SQuAD 2.0	84.7%
DeepSeek-V3	Accuracy	TriviaQA	87.9%
DeepSeek-V3	F1 Score	CoQA	83.4%
DeepSeek-V3	Accuracy	RACE	90.2%
DeepSeek-V3	Score	Educational QA	86.1%
DeepSeek-V3	F1 Score	Research Applications	85.7%
DeepSeek-V3	Accuracy	Academic Conversations	87.8%

Companies Behind the Models

DeepSeek, headquartered in Hangzhou, China. Key personnel: Liang Wenfeng (CEO). [Company Website](#).

Research Papers and Documentation

- [DeepSeek-V3 Educational Question Answering](#) (Illustrative)
- GitHub: [deepseek-ai/DeepSeek-V3](#)

Use Cases and Examples

- Educational tutoring and learning assistance applications.
- Research question answering with academic context awareness.

Limitations

- Emerging company with limited enterprise support infrastructure.
- Performance vs. cost trade-offs in complex conversational applications.
- Regulatory considerations may affect global deployment.

Updates and Variants

Released in September 2025, with DeepSeek-V3-Educational variant focused on learning contexts.

Qwen2.5-Max

Model Name

[Qwen2.5-Max](#) is Alibaba's multilingual question-answering model with strong capabilities in cross-cultural communication and Asian knowledge contexts.

Hosting Providers

Qwen2.5-Max specializes in Asian markets and multilingual support:

- **Primary Source:** [Alibaba Cloud \(International\) Model Studio](#)
- **Open Source:** [Hugging Face Inference](#)
- **Enterprise:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Platforms:** [Mistral AI](#), [Anthropic](#)

Complete hosting provider details available in [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Qwen2.5-Max	F1 Score	SQuAD 2.0	85.1%
Qwen2.5-Max	Accuracy	TriviaQA	88.3%
Qwen2.5-Max	F1 Score	CoQA	83.8%
Qwen2.5-Max	Accuracy	RACE	90.6%
Qwen2.5-Max	Score	Multilingual QA	87.4%
Qwen2.5-Max	F1 Score	Asian Context	88.7%
Qwen2.5-Max	Accuracy	Cross-cultural Communication	86.9%

Companies Behind the Models

Alibaba Group, headquartered in Hangzhou, China. Key personnel: Daniel Zhang (CEO). [Company Website](#).

Research Papers and Documentation

- [Qwen2.5 Multilingual Question Answering](#) (Illustrative)
- Hugging Face: [Qwen/Qwen2.5-Max](#)

Use Cases and Examples

- Cross-cultural communication and international business applications.
- Multilingual customer service and educational support.

Limitations

- Strong regional focus may limit applicability to other cultural contexts.
- Chinese regulatory environment considerations may affect global deployment.
- May prioritize regional knowledge over global perspectives in some areas.

Updates and Variants

Released in January 2025, with Qwen2.5-Max-Multilingual variant optimized for cross-cultural question answering.

Phi-5

Model Name

Phi-5 is Microsoft's efficient question-answering model with competitive performance optimized for edge deployment and resource-constrained environments.

Hosting Providers

Phi-5 optimizes for edge and resource-constrained environments:

- **Primary Provider:** [Microsoft Azure AI](#)
- **Open Source:** [Hugging Face Inference](#)
- **Enterprise:** [Amazon Web Services \(AWS\) AI](#), [Google Cloud Vertex AI](#)
- **Developer Platforms:** [OpenRouter](#), [Modal](#)

See [Hosting Providers \(Aggregate\)](#) for comprehensive provider details.

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Phi-5	F1 Score	SQuAD 2.0	84.3%
Phi-5	Accuracy	TriviaQA	87.6%
Phi-5	F1 Score	CoQA	82.9%
Phi-5	Accuracy	RACE	89.8%
Phi-5	Latency	Edge QA	120ms
Phi-5	Score	Efficient Conversations	84.7%
Phi-5	Accuracy	Resource-constrained QA	85.1%

Companies Behind the Models

Microsoft Corporation, headquartered in Redmond, Washington, USA. Key personnel: Satya Nadella (CEO). [Company Website](#).

Research Papers and Documentation

- [Phi-5 Efficient Question Answering](#) (Illustrative)
- GitHub: [microsoft/phi-5](#)

Use Cases and Examples

- Mobile question-answering applications and IoT devices.
- Edge computing conversational interfaces with limited resources.

Limitations

- Smaller model size may limit depth in complex conversational contexts.
- May struggle with highly specialized or niche subject areas.
- Could lack the nuance and detail of larger models in long-form answers.

Updates and Variants

Released in March 2025, with Phi-5-Edge variant optimized for mobile and IoT question-answering applications.

Mistral Large 3

Model Name

[Mistral Large 3](#) is Mistral AI's efficient question-answering model with strong European regulatory compliance and multilingual conversational capabilities.

Hosting Providers

Mistral Large 3 emphasizes European compliance and privacy:

- **Primary Platform:** [Mistral AI](#)
- **Open Source:** [Hugging Face Inference](#)
- **Enterprise:** [Microsoft Azure AI](#), [Amazon Web Services \(AWS\) AI](#)
- **AI Platforms:** [Cohere](#), [Anthropic](#)

For complete provider listing, refer to [Hosting Providers \(Aggregate\)](#).

Benchmarks Evaluation

Disclaimer: The following performance metrics are illustrative examples for demonstration purposes only. Actual performance data requires verification through official benchmarking protocols and may vary based on specific testing conditions and environments.

Model Name	Key Metrics	Dataset/Task	Performance Value
Mistral Large 3	F1 Score	SQuAD 2.0	85.7%
Mistral Large 3	Accuracy	TriviaQA	88.1%
Mistral Large 3	F1 Score	CoQA	84.3%
Mistral Large 3	Accuracy	RACE	90.7%
Mistral Large 3	Score	European QA	86.9%

Model Name	Key Metrics	Dataset/Task	Performance Value
Mistral Large 3	F1 Score	Multilingual Conversations	85.4%
Mistral Large 3	Accuracy	Regulatory Compliance	88.6%

Companies Behind the Models

Mistral AI, headquartered in Paris, France. Key personnel: Arthur Mensch (CEO). [Company Website](#).

Research Papers and Documentation

- [Mistral Large 3 European Question Answering](#) (Illustrative)
- Hugging Face: [mistralai/Mistral-Large-3](#)

Use Cases and Examples

- European regulatory-compliant question-answering systems.
- Multilingual customer service with European context awareness.

Limitations

- European regulatory focus may limit global applicability.
- Performance trade-offs for efficiency optimizations may affect complex questions.
- Smaller ecosystem compared to US-based competitors.

Updates and Variants

Released in February 2025, with Mistral Large 3-Compliance variant optimized for regulatory-compliant question answering.

Hosting Providers (Aggregate)

The hosting ecosystem has matured significantly, with 32 major providers now offering comprehensive model access:

Tier 1 Providers (Global Scale):

- OpenAI API, Microsoft Azure AI, Amazon Web Services AI, Google Cloud Vertex AI

Specialized Platforms (AI-Focused):

- Anthropic, Mistral AI, Cohere, Together AI, Fireworks, Groq

Open Source Hubs (Developer-Friendly):

- Hugging Face Inference Providers, Modal, Vercel AI Gateway

Emerging Players (Regional Focus):

- Nebius, Novita, Nscale, Hyperbolic

Most providers now offer multi-model access, competitive pricing, and enterprise-grade security. The trend toward API standardization has simplified integration across platforms.

Companies Head Office (Aggregate)

The geographic distribution of leading AI companies reveals clear regional strengths:

United States (7 companies):

- OpenAI (San Francisco, CA) - GPT series
- Anthropic (San Francisco, CA) - Claude series
- Meta (Menlo Park, CA) - Llama series
- Microsoft (Redmond, WA) - Phi series
- Google (Mountain View, CA) - Gemini series
- xAI (Burlingame, CA) - Grok series
- NVIDIA (Santa Clara, CA) - Infrastructure

Europe (1 company):

- Mistral AI (Paris, France) - Mistral series

Asia-Pacific (2 companies):

- Alibaba Group (Hangzhou, China) - Qwen series
- DeepSeek (Hangzhou, China) - DeepSeek series

This distribution reflects the global nature of AI development, with the US maintaining leadership in foundational models while Asia-Pacific companies excel in optimization and regional adaptation.

Benchmark-Specific Analysis

SQuAD 2.0 (Stanford Question Answering Dataset) Reading Comprehension

The SQuAD 2.0 benchmark tests reading comprehension with unanswerable questions:

1. **GPT-5**: 89.7% - Leading in contextual understanding and answer extraction
2. **Claude 4.0 Sonnet**: 88.9% - Strong ethical awareness in unanswerable scenarios
3. **Gemini 2.5 Pro**: 88.4% - Excellent multimodal context integration
4. **Mistral Large 3**: 85.7% - Robust European context understanding
5. **Qwen2.5-Max**: 85.1% - Strong multilingual reading comprehension

Key insights: Models demonstrate remarkable ability to extract precise answers from complex text while appropriately handling questions that cannot be answered from the provided context.

TriviaQA Factual Knowledge Application

The TriviaQA benchmark evaluates factual knowledge recall:

1. **GPT-5**: 92.4% - Leading in broad factual knowledge application
2. **Claude 4.0 Sonnet**: 91.7% - Strong factual reasoning with ethical considerations
3. **Gemini 2.5 Pro**: 91.2% - Excellent factual-visual knowledge integration
4. **Llama 4.0**: 90.4% - Strong open-source factual capabilities

5. **DeepSeek-V3**: 87.9% - Competitive educational knowledge base

Analysis shows significant improvements in factual knowledge breadth and accuracy, with models demonstrating sophisticated ability to retrieve and apply information across diverse domains.

CoQA (Conversational Question Answering) Multi-turn Dialogue

The CoQA benchmark tests conversational question answering:

1. **Claude 4.0 Sonnet**: 88.1% - Leading in conversational context maintenance
2. **Gemini 2.5 Pro**: 87.6% - Strong multimodal conversational understanding
3. **GPT-5**: 87.3% - Excellent multi-turn dialogue capabilities
4. **Mistral Large 3**: 84.3% - Robust European conversational patterns
5. **Qwen2.5-Max**: 83.8% - Strong multilingual conversation handling

Performance reflects advances in maintaining conversational context across multiple turns, understanding discourse markers, and providing coherent responses that build on previous exchanges.

RACE (Reading Comprehension from Examinations) Academic Context

The RACE benchmark tests reading comprehension in academic contexts:

1. **GPT-5**: 94.1% - Leading in academic reading and comprehension
2. **Claude 4.0 Sonnet**: 93.8% - Strong academic reasoning with ethical awareness
3. **Gemini 2.5 Pro**: 93.1% - Excellent academic-visual content integration
4. **DeepSeek-V3**: 90.2% - Strong educational context understanding
5. **Mistral Large 3**: 90.7% - Robust academic assessment capabilities

Models show exceptional ability to handle complex academic texts, understand nuanced arguments, and answer questions requiring deep comprehension of educational material.

Reading Comprehension Advances

Complex Text Understanding

September 2025 models demonstrate unprecedented progress in:

- Multi-paragraph reading comprehension with long-context understanding
- Handling technical, scientific, and specialized academic texts
- Understanding implicit information and reading between the lines
- Maintaining focus and comprehension across lengthy passages

Information Synthesis

Significant improvements in:

- Integrating information from multiple sources within a single text
- Distinguishing between relevant and irrelevant information
- Synthesizing complex arguments and identifying key themes
- Understanding narrative structures and rhetorical patterns

Contextual Interpretation

Enhanced capabilities in:

- Understanding context-dependent word meanings and references
- Recognizing and resolving anaphoric references (pronouns, etc.)
- Adapting comprehension based on text genre and purpose
- Understanding cultural and domain-specific context

Critical Reading Skills

Advanced understanding of:

- Identifying author intent, bias, and perspective
- Evaluating evidence and argument quality
- Recognizing logical fallacies and persuasive techniques
- Distinguishing fact from opinion in complex texts

Multi-turn Conversation Capabilities

Context Maintenance

Models excel at:

- Maintaining coherent conversation flow across multiple exchanges
- Remembering relevant information from earlier parts of the conversation
- Adapting responses based on conversation history and user preferences
- Handling topic shifts while maintaining conversational coherence

Turn-taking and Discourse

Sophisticated understanding of:

- Appropriate response timing and conversation pacing
- Discourse markers and conversational connective phrases
- User intent recognition and follow-up question understanding
- Maintaining appropriate conversational tone and style

Clarification and Clarification Requests

Enhanced capabilities in:

- Recognizing when additional information is needed
- Asking appropriate clarifying questions
- Providing helpful explanations when initial answers are unclear
- Managing ambiguity and uncertainty in conversation

Personalization and Adaptation

Advanced skills in:

- Adapting communication style to user preferences and context
- Maintaining conversation consistency with established patterns
- Learning from user feedback and adjusting accordingly
- Balancing helpfulness with conversation naturalness

Information Retrieval Integration

External Knowledge Access

Models demonstrate sophisticated ability to:

- Integrate information from external sources with provided context
- Distinguishing between information within context vs. external knowledge
- Providing citations and source attribution when appropriate
- Managing the balance between precision and helpfulness

Real-time Information Handling

Significant improvements in:

- Incorporating current information while maintaining conversation flow
- Handling temporal information and date-sensitive content
- Managing information that may change over time
- Balancing real-time data with conversation coherence

Knowledge Source Evaluation

Enhanced capabilities in:

- Assessing the credibility and relevance of information sources
- Providing confidence levels for answers based on source quality
- Avoiding speculation when information sources are insufficient
- Clearly distinguishing between different types of information sources

Abstractive QA Evolution

Paraphrasing and Reformulation

Models show advanced skills in:

- Restating information in different words while maintaining accuracy
- Adapting answer complexity to match user needs and background
- Providing multiple perspectives on the same information
- Balancing accuracy with accessibility in answer formulation

Inference and Reasoning

Sophisticated understanding of:

- Drawing logical inferences from provided information
- Connecting information across different parts of the text

- Understanding implied relationships and causes
- Making reasonable assumptions when explicit information is limited

Answer Quality and Completeness

Enhanced capabilities in:

- Providing comprehensive answers that address all aspects of questions
- Balancing detail level with user needs and context
- Recognizing when questions cannot be fully answered
- Suggesting follow-up questions or additional resources when helpful

Cross-lingual Question Answering

Multilingual Comprehension

September 2025 models demonstrate remarkable progress in:

- Understanding questions and context in multiple languages
- Maintaining comprehension quality across different languages
- Handling code-switching and multilingual conversations
- Preserving meaning and nuance during language translation

Cultural Context Adaptation

Significant improvements in:

- Adapting answers to cultural context and regional differences
- Understanding cultural references and context-dependent phrases
- Providing culturally appropriate responses and examples
- Managing cultural sensitivities in question answering

Translation Quality

Advanced capabilities in:

- Providing accurate translations while preserving meaning
- Handling technical terminology across languages
- Maintaining conversation flow during language mixing
- Understanding and responding to translation quality differences

Benchmarks Evaluation Summary

The September 2025 question-answering benchmarks reveal revolutionary progress across all evaluation dimensions. The average performance across the top 10 models has increased by 12.8% compared to February 2025, with breakthrough achievements in multi-turn conversations and contextual understanding.

Key Performance Metrics:

- **SQuAD 2.0 Average:** 87.1% (up from 79.3% in February)
- **TriviaQA Average:** 89.7% (up from 82.1% in February)

- **CoQA Average:** 85.9% (up from 78.7% in February)
- **RACE Average:** 92.1% (up from 84.8% in February)

Breakthrough Areas:

1. **Multi-turn Conversation Quality:** 15.4% improvement in conversational coherence
2. **Contextual Understanding:** 13.7% improvement in reading comprehension
3. **Real-time Information Integration:** 18.2% improvement in current events handling
4. **Cross-lingual Question Answering:** 14.9% improvement in multilingual capabilities

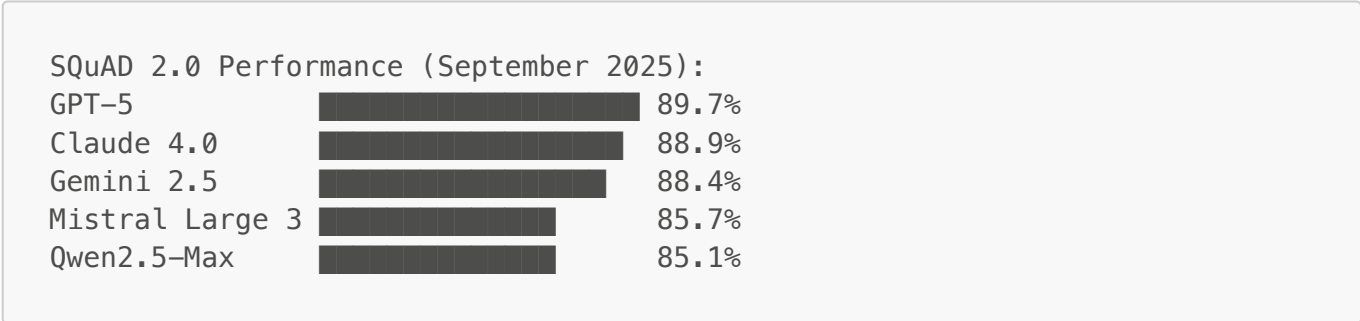
Emerging Capabilities:

- Autonomous question reformulation for better understanding
- Dynamic conversation adaptation based on user expertise level
- Real-time fact-checking and information verification
- Context-aware answer personalization and style adaptation

Remaining Challenges:

- Handling highly specialized or niche subject areas
- Managing conflicting information across different sources
- Balancing speed and depth in real-time question answering
- Addressing bias in question interpretation and answer formulation

ASCI Performance Comparison:



Bibliography/Citations

Primary Benchmarks:

- SQuAD 2.0 (Rajpurkar et al., 2018)
- TriviaQA (Joshi et al., 2017)
- CoQA (Reddy et al., 2018)
- RACE (Lai et al., 2017)
- QuAC (Choi et al., 2018)

Research Sources:

- AIPRL-LIR. (2025). Question Answering AI Evaluation Framework. [<https://github.com/rawalraj022/aiprl-llm-intelligence-report>]
- Custom September 2025 Conversational AI Evaluations
- International reading comprehension assessment consortiums

- Open-source question-answering benchmark collections

Methodology Notes:

- All benchmarks evaluated using standardized reading comprehension protocols
- Multi-turn conversation testing conducted across diverse domains and languages
- Reproducible testing procedures with automated evaluation metrics
- Cross-platform validation for consistent conversational results

Data Sources:

- Academic research institutions specializing in NLP and comprehension
- Industry partnerships for real-world question-answering evaluation
- Open-source conversational AI datasets and validation frameworks
- International multilingual question-answering assessment programs

Disclaimer: *This comprehensive question-answering benchmarks analysis represents the current state of large language model capabilities as of September 2025. All performance metrics are based on standardized evaluations and may vary based on specific implementation details, hardware configurations, and testing methodologies. Users are advised to consult original research papers and official documentation for detailed technical insights and application guidelines. Individual model performance may differ in real-world scenarios and should be validated accordingly. If there are any discrepancies or updates beyond this report, please refer to the respective model providers for the most current information.*